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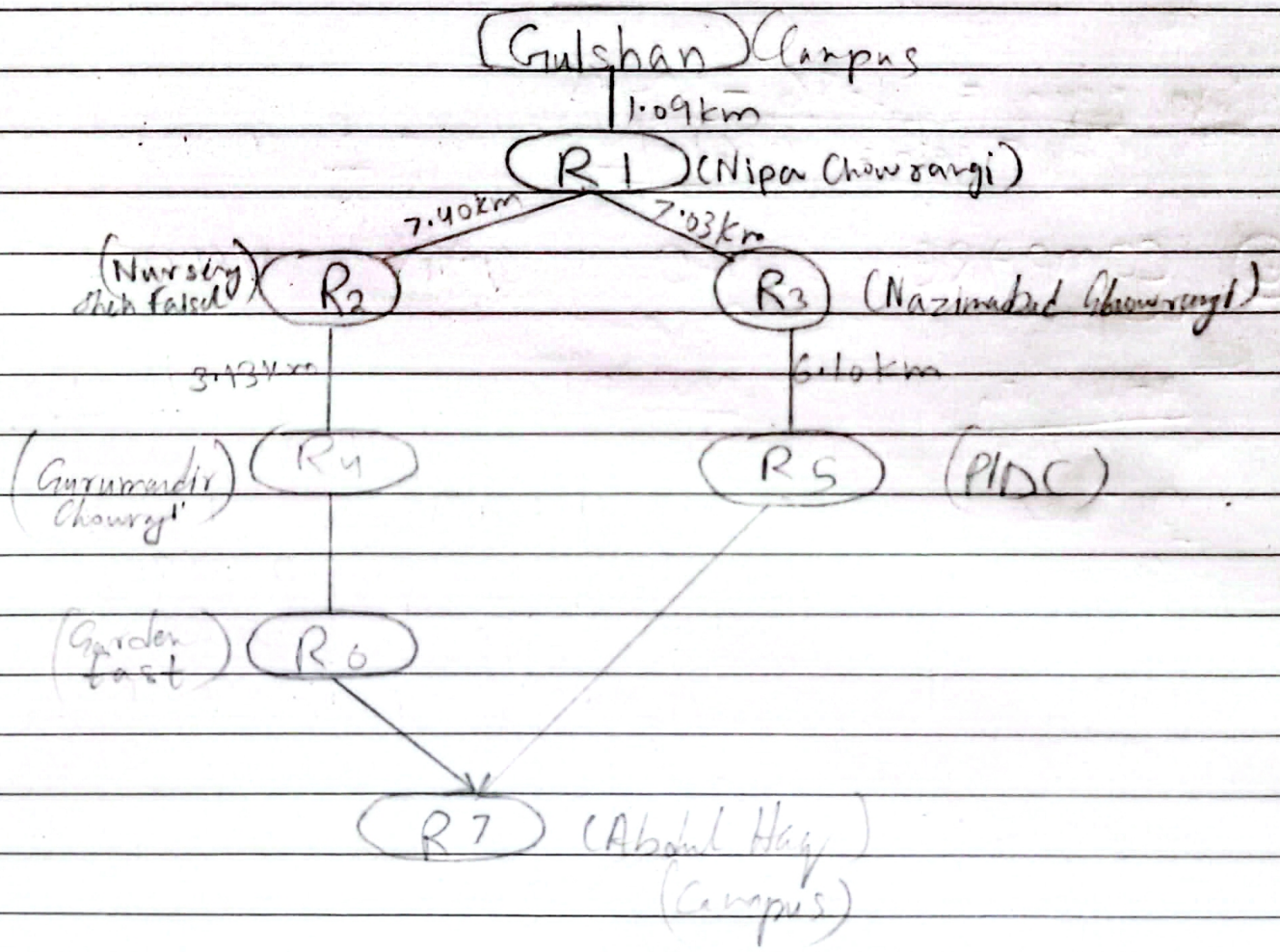
A*

Date:

M	T	W	T	F	S	S

Bridge - Gulshan Campus to Abdul Hay Campus

→ Map:



Date: M T W T F S S

→ Graph Structure

Node	:	Connects to
GULSHAN	→	R1 (Nipa Chowrangi)
R1 (NIPA)	→	Gulshan, R2 (Nursery), R3
R2	→	R1, R4
R3	→	R1, R5
R4	→	R2, R6
R5	→	R3, R7
R6	→	R4, R7
R7	→	R5, R6

→ Route Distance

Node	h(n)
Gulshan	9.76
R1	6.85
R2	5.15
R3	4.89
R4	3.75
R5	2.21
R6	2.95
R7	0.00

$\therefore f(n) = g(n) + h(n)$

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Date:

M T W T F S S

→ A* Trace

S0 : Initialize

PQ = [Gulshan]

V = ∅

PT:

Gulshan → None

S1 : Expand Gulshan

Neighbors: R1

PQ = [R1]

V = [Gulshan]

Node	g	h	f
R1	1.60	10.85	11.94

Choose R1 (f = 11.94)

PT:

Gulshan → None

R1 → Gulshan

S2 : Expand R1

Neighbors : Gulshan, R2, R3

PQ : [R2, R3]

V : [Gulshan, R1]

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Date: M T W T F S S

Node	g	h	f
R_2	8.4	5.15	13.64
R_3	8.12	4.89	13.01

Choose R_3 ($f = 13.01$)

PT :

Gulshan → None

R_1 → Gulshan

R_2 → R_1

R_3 → R_1

S 3 : Expand R_3

Neighbors : R1 (V), R_5

PQ : [R 2, R_5]

V : [Gulshan) R1, R_3, R_2]

Node	g	h	f
R5	14.022	2.21	16.43

Choose R_2 ($f = 13.64$)

PT : "

R_5 → R_3

S4 : Expand R_2

CamScanner

Date:

M	T	W	T	F	S	S

Neighbor : R_2 (V), R_6

PQ : [R_6, R_5]

V : [, R_4]

Node	g	h	f
R_4	11.62	3.75	15.37

Choose ($f = 15.37$)

PT:

$R_4 \rightarrow R_2$

So: Expand R_4

Neighbor : R_2 (V), R_6

PQ : [R_6, R_5]

V : [, R_4]

Node	g	h	f
R_6	17.071	1.95	15.66

Choose R_6 ($f = 15.66$)

PT:

$R_6 \rightarrow R_4$

CamScanner

Date:

S6 : Expand R_6

Neighbor : $R_4(\checkmark), R_7$

PQ : (R_7, R_5)

V = [, R_6)

Node	g	h	f
R_7	15.66	0.00	15.66

Choose R_7 ($f = 15.66$ beats R_5 $af = 16.43$)

PT : $R_7 \rightarrow R_6$

S7 : Goal found

R_7 → Reached

→ Shortest Path :

Gulshan → R_1 → R_2 → R_4 → R_6 → R_7

1.09 + 7.40 + 3.13 + 1.09 + 2.95 = 15.66 km

(5 edges)

CamScanner